TELECOM ANALYTICS

# Customer Churn Prediction

Using usage, plan, complaint & billing data to find at-risk customers and power proactive retention

4

Data Domains

AI

Risk Scoring

Churn

Retention Impact

Prepared for Retention Strategy Review

# The Business Problem

Churn quietly erodes telecom revenue — and it's cheaper to prevent than to replace

**5–7x** More expensive to acquire a new customer than retain one

**15–25%** Typical annual churn rate in competitive telecom markets

**70%+** Of churn shows early warning signals in usage & billing data

## Why Predict Churn?

- Spot at-risk customers weeks before they cancel
- Target retention offers only where they're needed — protect margins
- Turn complaint & billing friction into fixable root causes
- Prioritize high-value accounts for proactive outreach
- Track intervention success with a measurable risk score

# Four Data Domains, One Risk Signal

Each customer's digital footprint is combined into a unified churn-risk profile

## Usage Data

- Call minutes & data volume trend
- Declining engagement patterns
- Device / network switch events

## Plan Data

- Current plan & tenure
- Downgrade / contract type
- Add-ons and family lines

## Complaint Data

- Support ticket frequency
- Unresolved issue count
- Sentiment of interactions

## Billing Data

- Late or missed payments
- Bill shock / price increases
- Refunds and credit history

# How the Prediction Works

A five-stage pipeline turns raw records into a ranked, actionable risk list

01

### Collect & Merge

Unify usage, plan, complaint, and billing records per customer

02

### Feature Engineering

Derive trends: usage drop-off, ticket frequency, payment delays

03

### Model Training

Train classifiers (e.g., gradient boosting / logistic regression) on historical churners

04

### Risk Scoring

Score every active customer 0–100 on churn probability

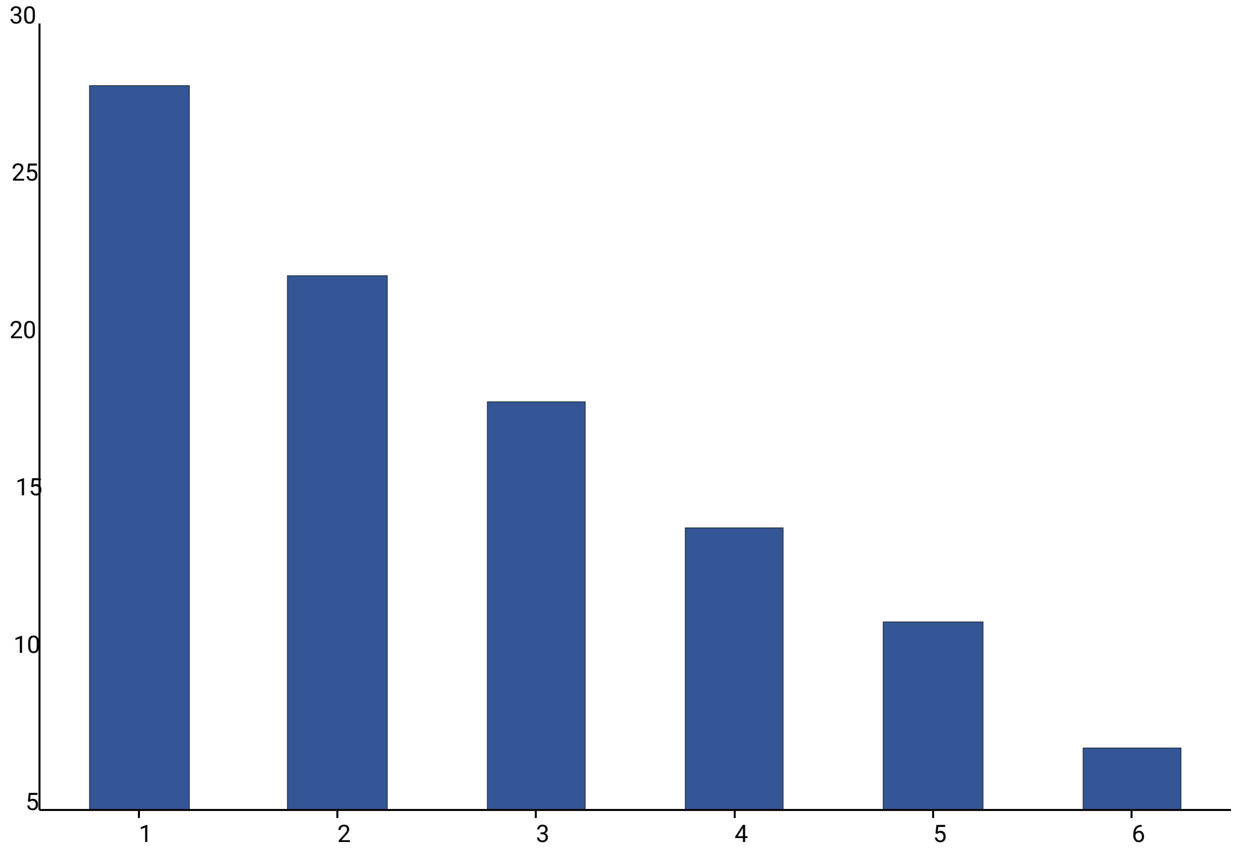
05

### Retention Action

Route high-risk, high-value customers to targeted campaigns

# Top Churn Risk Indicators

Relative importance of each signal in flagging at-risk customers




## What Drives Risk

- Usage decline is the strongest single predictor — a 30%+ drop over 60 days is a major flag
- Two or more complaints in 90 days sharply raises risk
- Payment delays often precede cancellation by 1–2 billing cycles
- Customers nearing contract end are 3x more likely to shop competitors

# From Score to Segment

Every customer is scored, then grouped into a clear action tier

62%

## Low Risk

Stable usage, on-time billing, no recent complaints. Standard engagement only.

24%

## Watch List

One or two mild signals (slight usage dip or a single complaint). Monitor monthly.

14%

## High Risk

Multiple strong signals — usage drop, complaints, or payment delays. Immediate outreach.

### Customer Base Composition


Model output re-scored monthly as new usage, billing, and complaint data arrives.

# Retention Strategies by Segment

Match the offer to the risk driver — not every at-risk customer needs a discount

### Fix the Root Cause

Fast-track unresolved complaints; proactive call before contract renewal

### Targeted Incentives

Loyalty discounts or data top-ups for high-value, high-risk accounts only

### Plan Right-Sizing

Recommend a better-fit plan before customers downgrade or leave entirely

### Billing Transparency

Alert & explain bill increases early; offer flexible payment options

### Proactive Outreach

Personal check-in calls for the top decile of churn-risk scores

### Win-Back Journeys

Automated re-engagement sequences triggered the moment risk score spikes